

Mathematical Preliminaries

Matrices

A matrix is a rectangular array of numbers arranged in rows and columns. The size of a matrix is determined by the number of rows and columns it contains. Matrices are widely used in mathematics, physics, engineering, computer science, and many other fields.

Matrix Addition

Matrix addition is an operation that involves adding the corresponding elements of two matrices of the same size. Mathematically, if A and B are both $m \times n$ matrices (with m rows and n columns), then their sum, denoted as C , is an $m \times n$ matrix where the entry in the i -th row and j -th column of C is given by:

$$C_{ij} = A_{ij} + B_{ij}$$

In this formula, A_{ij} is the entry in the i -th row and j -th column of matrix A , and B_{ij} is the entry in the i -th row and j -th column of matrix B . Note that in order to add two matrices, they must be of the same size, i.e., they must have the same number of rows and the same number of columns.

In our scripts, matrix addition will be denoted $C = A + B$. Any number of spaces may appear before or after symbols and matrix names in our formulas.

Matrix Multiplication

Matrix multiplication is a way of combining two matrices to produce a new matrix. The resulting matrix has the same number of rows as the first matrix and the same number of columns as the second matrix.

Mathematically, if A is an $m \times n$ matrix and B is an $n \times p$ matrix, then the product of A and B , denoted as C , is an $m \times p$ matrix where the entry in the i -th row and j -th column of C is given by:

$$C_{ij} = \sum_{k=1}^n (A_{ik} B_{kj})$$

In this formula, A_{ik} is the entry in the i -th row and k -th column of A , and B_{kj} is the entry in the k -th row and j -th column of B .

In our scripts, matrix multiplication will be denoted $C = A * B$.

Matrix Transpose

Matrix transpose is an operation that involves flipping a matrix over its diagonal. Mathematically, if A is an $m \times n$ matrix, then its transpose, denoted as A^T , is an $n \times m$ matrix where the entry in the i -th row and j -th column of A^T is given by:

$$(A^T)_{ij} = A_{ji}$$

In other words, the rows of the original matrix become columns in the transposed matrix, and the columns of the original matrix become rows in the transposed matrix.

In our scripts, matrix transpose will be denoted $B = A'$ (Single quotation mark).

Script Format

Your program will process a script containing commands to define matrices and perform operations on them. Every line of a script has one of two forms:

- a definition of a new matrix from a list of values
- a definition of a new matrix created from a formula containing at least one operator.

A new matrix is defined by providing:

- a single uppercase character
- zero or more spaces
- an equals symbol
- zero or more spaces
- a positive integer containing one or more digits that gives the number of rows of the matrix (call this number NR)
- one or more spaces
- a positive integer containing one or more digits that gives the number of columns of the matrix (call this number NC)
- zero or more spaces
- a left square bracket
- zero or more spaces
- NC integers separated by spaces and terminated by a semicolon. There might or might not be spaces surrounding the semicolon.
- NR-1 additional lists of NC integers as described in the previous item (total NR lists of NC integers)
- a right square bracket

- zero or more spaces
- a newline

Example: $A = \begin{bmatrix} 3 & 2 \\ 4 & 5 \end{bmatrix}$; $\begin{bmatrix} 19 & -34 \\ 192 & -9110 \end{bmatrix}$; $\backslash n$

A formula is defined by providing:

- a single uppercase character that has not appeared in the script on a previous line
- zero or more spaces
- an equals symbol
- zero or more spaces
- an expression consisting of + (for addition), * (for multiplication), ' (for transpose), left and right parentheses, and matrix names. The precedence of the operators, from highest to lowest, is:
 - transpose
 - multiplication
 - addition

Parentheses can be used to change the order of evaluation. An expression may contain spaces anywhere throughout the expression around any of the symbols.

- a newline

Example: $Z = (A + B)' * C * (D' + A)'$ $\backslash n$

Once a symbol has been defined, it cannot be redefined (i.e., reassigned). A symbol can appear multiple times on the right-hand side of a formula.

Scripts will always be syntactically valid. Extra spaces (or no spaces) may surround the *tokens* (e.g., matrix name, equals sign, left and right square brackets, left and right parentheses, semicolons, operators, numbers). When two integers are next to each other, they will be separated by at least one space, or, in the case where we have reached the end of a row, a semicolon. There might or might not be spaces around the semicolons.

Scripts will always be semantically valid, meaning that only valid formulas will be provided(input is guaranteed valid). For instance, a script will not contain formulas that cannot be evaluated for matrices of the given dimensions.

To summarize: your main task in parsing the text file is NOT to worry about error cases because there will be no error cases. Focus on implementing the order of operations correctly and evaluating the expressions correctly.

Data Structures

matrix_sf

The following data type is defined in hw7.h, which you will use to implement the functions in this assignment. As its name and members suggest, it represents a matrix of integers:

```
typedef struct {
    char name;
    unsigned int num_rows;
    unsigned int num_cols;
    int values[]; // stores the matrix's contents in row-major order
} matrix_sf;
```

To allocate the memory for a matrix, make a call to malloc:

```
matrix_sf *m = malloc(sizeof(matrix_sf)+num_rows*num_cols*sizeof(int));
```

Then you need to assign values to the name, num_rows, num_cols, and values members of the struct.

To deallocate the memory for a matrix pointed to by m, make a call

to free: free(m);

Do NOT attempt to separately free() the values array because it is contiguous with the other members of the struct.

bst_sf

The following data type is also defined in hw7.h and represents a node in a [binary search tree](#), where the tree uses the names of the matrices to sort the nodes. Recall that matrix names are unique.

```
typedef struct bst_sf {
    matrix_sf* mat;
    struct bst_sf* left_child;
    struct bst_sf* right_child;
} bst_sf;
```

This is a simple BST and is not balanced. Don't get fancy or cute here, or else your code will fail many unit tests. Implement a basic, simple BST. `left_child/right_child` is NULL if a node has no left/right child.

We will use a BST as a symbol table to associate names of matrices appearing in formulas with `matrix_sf` objects. This is explained in more detail later.

Functions to Implement

You'll most likely want to implement these in the order listed, but jump around if that works better for your workflow.

```
matrix_sf* add_mats_sf(const matrix_sf *mat1, const matrix_sf *mat2)
```

- Perform the matrix addition `mat1+mat2` and return the sum.

```
matrix_sf* mult_mats_sf(const matrix_sf *mat1,
                       const matrix_sf *mat2)
```

- Perform the matrix multiplication `mat1*mat2` and return the product.

```
matrix_sf* transpose_mat_sf(const matrix_sf *mat)
```

- Return the transpose of `mat`.

```
matrix_sf* create_matrix_sf(char name, const char *expr)
```

- Parse a string (`expr`) containing a valid definition of a new matrix (as described in the "Script Format" section) and return a pointer to a correctly initialized `matrix_sf` struct. For this function, `expr` does not necessarily end with a newline character. The spacing between tokens in `expr` can vary significantly inside the string.
- For example, for `name = 'G'`, `expr = "2 5 [ 8 4 9 1 13 ; -5 0 6 22 24 ;]"`, the return value would point to a `matrix_sf` struct with the following contents:
`name = 'G'`
`num_rows = 2`
`num_cols = 5`
`values = {8, 4, 9, 1, 13, -5, 0, 6, 22, 24}`

```
bst_sf* insert_bst_sf(matrix_sf *mat, bst_sf *root)
```

- Given a pointer to the bst_sf struct root, which could be NULL, insert the provided matrix mat into the BST without making a copy of mat. The function must ensure that the sorted property of the BST is maintained. The function creates a new BST if root is NULL.
- Return a pointer to the root of the updated (or new) BST.
- Assume that no other matrix_sf exists in the BST that has the same name as mat.
- The function should perform no validation whatsoever of the contents of mat. You may assume that mat != NULL and that no other matrix_sf is stored in the BST with the same value for mat->name. The function must insert mat into the BST even if any of num_rows, num_cols, or values is invalid.

```
matrix_sf* find_bst_sf(char name, bst_sf *root);
```

- Given a pointer to the bst_sf struct root, which could be NULL, locate and return a pointer to a matrix named name, if it exists.
- If no matrix of the given name is in the BST, return NULL.
- Assume that at most one matrix named name exists in the BST.
- Assume that the root points to a valid BST or is NULL.
- Assume that no two BST nodes point to the same matrix_sf struct.

```
void free_bst_sf(bst_sf *root);
```

- Given a pointer to the bst_sf struct root, which could be NULL, free all the nodes of the tree.
- Also, free all the matrix_sf structs pointed to by the BST nodes.
- Assume that the root points to a valid BST or is NULL.
- Assume that no two BST nodes point to the same matrix_sf struct.

```
char* infix2postfix_sf(char *infix);
```

- Given an infix expression infix, convert it to its equivalent postfix expression. Return the newly allocated string containing the postfix expression.
- You are strongly encouraged to adapt the stack-based algorithm you learned in CSE 214/260 to [convert an infix expression into postfix](#). You'll need to handle +, *, ', (,) and matrix names. (Note that ' is a unary

operator and has higher precedence than + or *.)

- Do not forget to consider the precedence of the operators for this conversion.
- Any memory allocated by the function must be deallocated before returning, except for the string that contains the postfix expression.

```
matrix_sf* evaluate_expr_sf(char name, char *expr, bst_sf *root);
```

- Given the name of a new matrix (name), an expression involving one or more matrices and one or more matrix operations (expr), and a pointer to the root of a BST (root), perform the following tasks:
 - Evaluate expr, storing the resulting matrix in a new matrix called name.
 - Return a pointer to the new matrix.
- Any memory allocated by the function must be deallocated before returning, except for the matrix that contains the value of the expression.
- Some considerations:
 - expr is an infix expression. Use the function infix2postfix_sf to convert it to postfix.
 - To help you evaluate the postfix expression, create a stack of pointers to matrix_sf structs. When you encounter a +, *, or ' in the expression, pop one (for ') or two (for +, *) matrices off the stack and perform the operation. As you evaluate a +, *, or ' operation, create a new matrix_sf and push it onto the stack. Do NOT add the new matrix to the BST.
 - Remember: the function must deallocate the postfix expression string returned by calling the infix2postfix_sf function.
 - Remember: the function must deallocate any memory it allocates while evaluating the expression. (Again, the returned matrix is an exception.) When you need to create a new matrix on-the-fly, give it a non-alphabetical name. Later on, when this matrix is popped off the stack to apply an operator (+, *, '), you can easily identify it as a matrix that evaluate_expr_sf needed to create. You can then safely free() it after applying the operator and getting the result. Once you start implementing this function, you will understand this advice better.

Example: Suppose expr = "P'*K+A".

1. First transform this expression into its postfix format, which is "P'K*A+", by calling the function infix2postfix_sf.
2. Initialize an empty stack of pointers to matrix_sf structs.
3. Start processing characters sequentially from "P'K*A+".
4. We see P in the input. Push the matrix called P on the stack. (Call find_bst_sf('P', root) to get a pointer to that matrix.)
5. We see ' in the input. Transpose is a unary operator. Pop the top element of the stack, create a new matrix that stores the transpose, and push

- that new matrix onto the stack. Do NOT add the new matrix to the BST.
6. We see K in the input. Push the matrix called K on the stack.
 7. We see * in the input. Pop the top two elements off the stack, create a new matrix that stores the product, and push that new matrix onto the stack. If either or both of the popped matrices do NOT have an alphabetical name (oh look at that, one of them doesn't), free() the applicable matrix/matrices.
 8. We see A in the input. Push the matrix called A on the stack.
 9. We see + in the input. Pop the top two elements off the stack, create a new matrix that stores the sum, and push that new matrix onto the stack. If either or both of the popped matrices do not have an alphabetical name, free() the applicable matrix/matrices.
 10. We have reached the end of the input. We will return a pointer to the matrix on top of the stack. But first, as necessary, free() your stack data structure and any other memory you allocated but haven't deallocated yet.

```
matrix_sf* execute_script_sf(char *filename);
```

- Given the name of a file containing a script as described under “Script Format” above, execute the contents of the file and return a pointer to the final, named matrix created on the last line of the script. For example, suppose “C = A*B + D” is the last line of the file. The function will return a pointer to the matrix_sf struct named 'C'.
- You are encouraged to use the function [getline](https://man7.org/linux/man-pages/man3/getline.3.html), which can read a line of text from a file and will even allocate the memory to store the string. (You'll have to deallocate the memory yourself.) You would call it as follows:

```
char *str = NULL; // Read https://man7.org/linux/man-pages/man3/getline.3.html
FILE *file = ...;
size_t max_line_size = MAX_LINE_LEN; // defined in hw7.h
getline(&str, &max_line_size, file);
```

- The function will need to call some other functions as needed:
 - create_matrix_sf when a line in the script defines a new matrix by providing its contents (e.g., A = 3 4 [...]), followed by a call to insert_bst_sf to insert the new matrix into the BST
 - evaluate_expr_sf when a line in the script creates a new matrix by performing one or more matrix operations, followed by a call to insert_bst_sf to insert the new matrix into the BST

Testing & Grading Notes

- During grading, only your hw7.c file will be copied into the grading framework's directory for processing. Make sure all of your code is self-contained in that file.
- Most test cases will be executed twice to check (a) the correctness of the return value from a function; and (b) the correctness of your memory usage by Valgrind.

See unit_tests.c for details. Credit will be awarded for a successful Valgrind test only when its paired computational test passes.

- It is useful and good practice to create helper functions as you are working on the assignment. These are allowed and will not interfere with grading. However, make sure to only modify hw7.c.

Running and Testing Your Code:

1. You will provide enough (at least 10) test cases of your implementation as well, which show that it takes care of most of the cases

Grading Notes

- The TAs will test your code on a Linux VM using their own tests. Marks depend on how many test cases (yours and the TAs' test cases) your code passes. 100% pass ensures 25 marks for the Testing part. 5 points are reserved for software engineering practice.
- The submitted code will be checked for plagiarism using tools. Any instances of plagiarism will result in zero marks, and the case will be forwarded to the department for disciplinary action.
- You will receive zero marks if the TAs cannot compile your code on their machine using the makefile.
- Any warnings in your code indicate potential bugs. 50% of the marks will be deducted.